Digital Logic Design



Submitted To: Sir Kashif
Submitted By: Usman, Abdullah Satti, Farhan
Enrolment: 037, 048, 012

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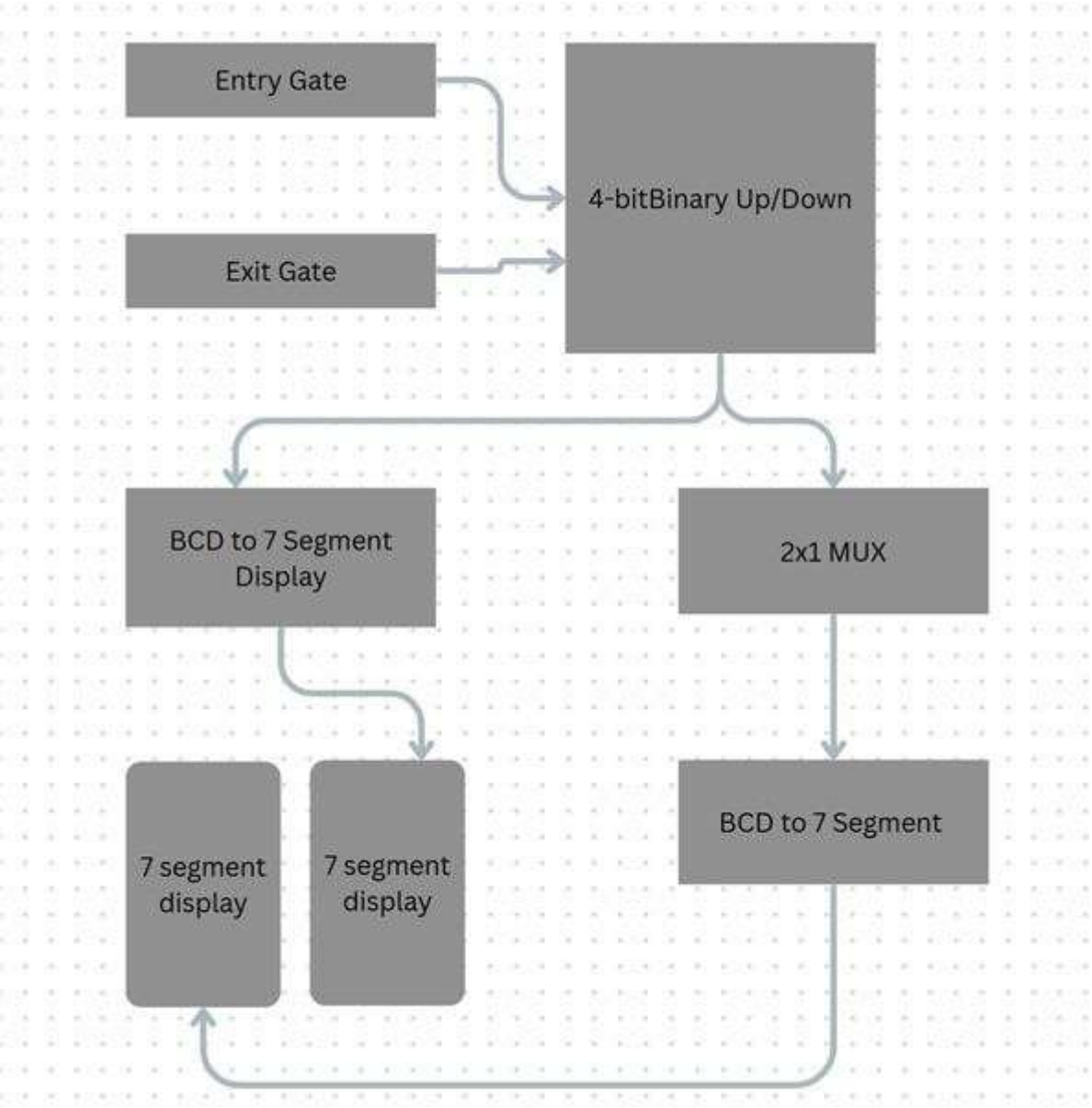
Project Title: Smart Parking Lot Management System

1. System Overview

The Smart Parking Lot Management System is a digital logic-based solution designed to automate the monitoring and management of a parking facility. Using basic logic gates and sequential circuits, the system:

- Detects the presence of vehicles at the entry and exit points.
- Allows vehicle entry only when parking space is available.
- Updates the internal binary counter representing the current number of parked vehicles.
- Prevents the entry of additional vehicles when the parking lot is full.
- Prevents vehicle exit when no cars are parked.
- Displays the current count of parked vehicles both in binary and on a 7-segment display.
- Includes an administrative manual override/reset feature to reinitialize the system.

2. Block Diagram



Admin Override/Reset Switch ---> System Reset & Control Logic

Explanation:

- The entry and exit sensors are implemented using SPST (single pole single throw) switches in simulation.
- Logic gates ensure that the vehicle can only enter if the count is less than the maximum (e.g., 15 slots).
- On each valid entry or exit, a rising-edge signal triggers the up/down counter.
- The current count is displayed using a 4-bit binary output (for circuit debugging) and a 7-segment display (for user readability).

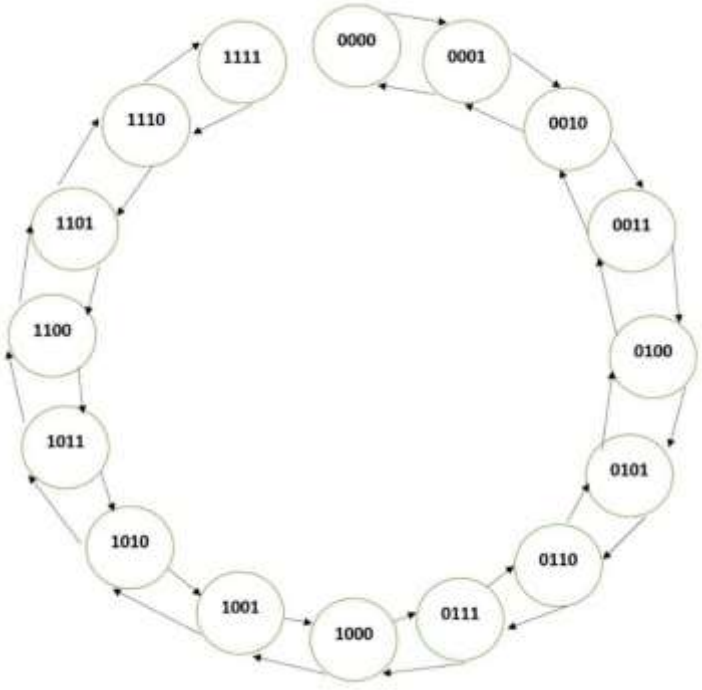
- A reset signal can bring the entire system back to its initial state.

3. Logic Components Used

- **Switches (Sensors)**: Represent presence of vehicles at entry and exit points.
- **AND, OR, NOT Gates**: Used to create the logical conditions for valid entry/exit, full detection, and empty detection.
- **JK Flip-Flops (7473)**: Used to build a 4-bit synchronous up/down counter. Toggled on valid entry or exit signals.
- **MUX IC 74157**: Used for data selection logic to simplify control between manual reset and sensor-driven operations.
- **BCD to 7-Segment Decoder (7448)**: Converts binary counter output into a readable 7-segment display format.
- **LED Indicators**: Indicate parking full status, entry/exit gate operation, and system availability.
- **Manual Reset Button**: Admin control for system override or reset.

5. State Diagram

The counter maintains a state representing the number of parked vehicles:



- **Upward Transition**: Occurs on valid entry.
- **Downward Transition**: Occurs on valid exit.

- **Blocked Transitions**:
  - No increment when count is 15 (FULL). ○ No decrement when count is 0 (EMPTY).

4. Truth Tables

Entry Control Logic

Entry Sensor	Parking Full	Allow Entry	Entry Gate Signal
0	X	0	0
1	0	1	1
1	1	0	0

Exit Control Logic

Exit Sensor	Parking Empty	Allow Exit	Exit Gate Signal
0	X	0	0
1	0	1	1
1	1	0	0

To complete your report with detailed 4-bit Up/Down Counter Truth Table and Excitation Table (for JK Flip-Flops), we'll include the current state, next state, and the required J and K inputs for each flip-flop (Q_0 to Q_3). Since your counter:

- **Counts up from 0000 to 1111, and stops at 1111.**
- **Counts down from 1111 to 0000, and stops at 0000.**
- **Does not roll over (e.g., does not go from 1111 → 0000 in up mode).**

We assume it's a controlled counter, enabled only when entry or exit is valid.
Flip-Flop Excitation Table (JK)

Current (Q_n)	Next (Q_{n+1})	Jn	Kn
0	**0**	**0**	**X**

Current (Qn) Next (Q_{n+1}) Jn Kn

0 1 1 X

1 0 X 1

1 1 X 0

- **Current State (Q_3 Q_2 Q_1 Q_0)**
- **Next State for Up and Down**
- **Required JK inputs for each flip-flop (J_3–J_0, K_3–K_0)**
- **For both Up Counter and Down Counter**
- **Stops at 1111 in Up and at 0000 in Down**

Combined Truth and Excitation Table for 4-Bit Up/Down Counter (Stops at Limits)

Current State (Q_3 Q_2 Q_1 Q_0)	Next State (UP)	Next State (DOWN)	J_3 K_3 J_2 K_2 J_1 K_1 J_0 K_0
0000	0001	0000	0 X 0 X 0 X 1 X
0001	0010	0000	0 X 0 X 1 X X 1
0010	0011	0001	0 X 0 X X 1 1 X
0011	0100	0010	0 X 1 X X 1 X 1
0100	0101	0011	0 X X 1 0 X 1 X
0101	0110	0100	0 X X 1 1 X X 1
0110	0111	0101	0 X X 1 X 1 1 X
0111	1000	0110	1 X 0 X 0 X X 1
1000	1001	0111	X 1 0 X 1 X 1 X
1001	1010	1000	X 1 0 X X 1 X 1
1010	1011	1001	X 1 0 X 1 X 1 X
1011	1100	1010	X 1 1 X 0 X X 1
Current State (Q_3 Q_2 Q_1 Q_0)	Next State (UP)	Next State (DOWN)	J_3 K_3 J_2 K_2 J_1 K_1 J_0 K_0
1100	1101	1011	X 1 X 1 1 X 1 X

1101	1110	1100	X 1 X 1 X 1 X 1
1110	1111	1101	X 1 X 1 1 X 1 X
1111	1111	1110	0 X 0 X 0 X X 1

- X means "don't care" (either 0 or 1, depending on implementation).
- Flip-Flops are toggled only if:
  - J = 1 and K = X → Flip from 0 to 1
  - J = X and K = 1 → Flip from 1 to 0
- The counter stops at 1111 while counting up and at 0000 while counting down (no rollover).

Final Simplified K-Map Expressions (Up Counter):

Flip-Flop	J Expression	K Expression
Q_0	1	1
Q_1	Q_0	Q_0
Q_2	$Q_1 \cdot Q_0$	$Q_1 \cdot Q_0$
Q_3	$Q_2 \cdot Q_1 \cdot Q_0$	$Q_2 \cdot Q_1 \cdot Q_0$

Down Counter

Just flip the conditions:

- Q_1 toggles when $Q_0 = 0$
- Q_2 toggles when $Q_1 = 0$ and $Q_0 = 0$
- Q_3 toggles when $Q_2 = 0$, $Q_1 = 0$, $Q_0 = 0$

Flip-Flop	J (Down)	K (Down)
Q_0	1	1

Flip-Flop	J (Down)	K (Down)
Q_1	Q_0'	Q_0'
Q_2	$Q_1' \cdot Q_0'$	$Q_1' \cdot Q_0'$

Q_3 $Q_2' \cdot Q_1' \cdot Q_0'$ $Q_2' \cdot Q_1' \cdot Q_0'$

6. K-Map Simplification Examples

A. Full Detection Logic

Let A, B, C, D be outputs of the counter:

- Full condition: When A=1, B=1, C=1, D=1 (Binary 1111 = Decimal 15)
- Boolean Equation: **FULL = A · B · C · D**

B. Empty Detection Logic

- Empty condition: A=0, B=0, C=0, D=0 (Binary 0000 = Decimal 0)
- Boolean Equation: **EMPTY = A' · B' · C' · D'**

These equations are simplified from 4-variable K-Maps using SOP (Sum of Products) method.

7. Circuit Diagram

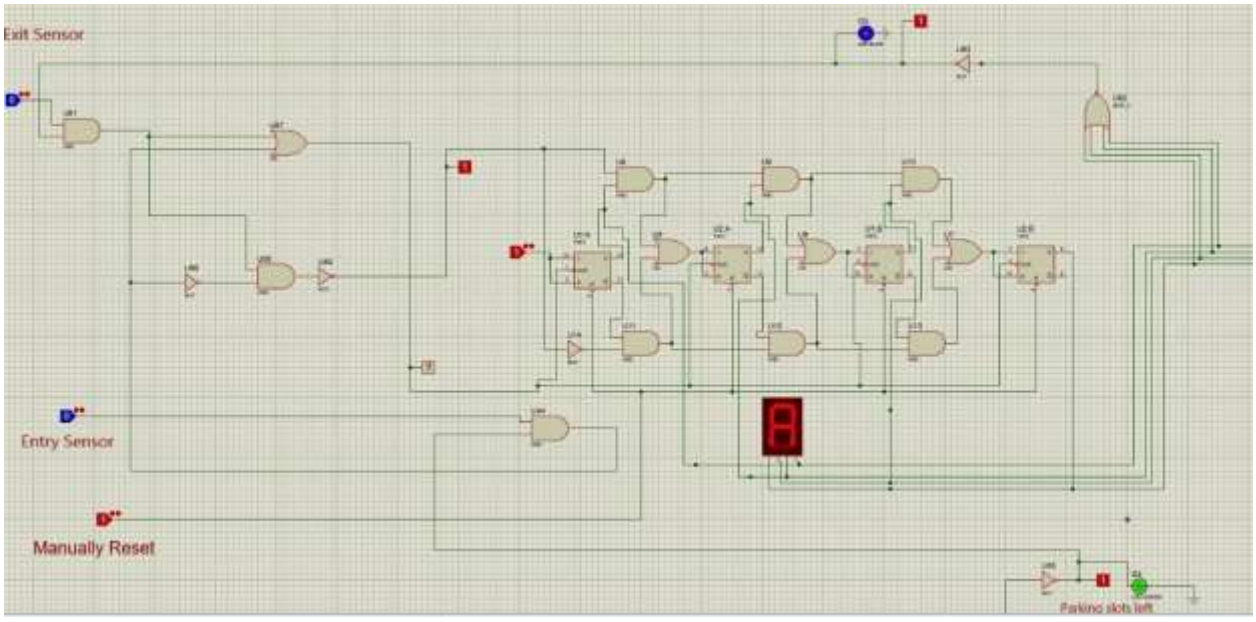


Figure: Proteus Simulation of Smart Parking Lot Controller

Explanation of Diagram:

- **Switches** at the bottom-left simulate the entry, exit, and manual reset sensors.
- The **AND/OR/NOT gates** form the decision logic for enabling the clock signal to the counter.

- The **flip-flops** configured as an up/down counter (likely 7473 ICs) track vehicle entries and exits.
- The **MUX IC 74157** handles conditional input routing.
- The **7448 Decoder** converts the 4-bit binary count into a 7-segment display format.
- The **7-segment display** shows the number of vehicles parked.
- The **LEDs** indicate gate status and parking availability (FULL signal).

8. Design Summary

Feature	Implementation Details
Entry/Exit Detection	Switches simulating sensors
Space Counter	4-bit Up/Down Counter using JK Flip-Flops (7473)
Display	Binary Output + 7-Segment Display (using 7448 BCD decoder)
Entry Control Logic	Prevents entry if parking is FULL
Exit Control Logic	Prevents exit if parking is EMPTY
Admin Override/Reset	Switch input resets the counter
MUX Logic Control	74157 MUX used for managing sensor vs manual logic

Circuit Simulated Using Proteus Design Suite

Conclusion:

The Smart Parking Lot Management System is a digital logic-based solution that efficiently manages parking facilities by detecting vehicle presence at entry and exit points, updating parking counts in real-time, preventing overcrowding and unauthorized exits, and displaying current parking counts on a 7-segment display, with admin override/reset functionality for enhanced control.